

Lalit Manam

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Research Interests

3D Computer Vision, Structure-from-Motion, Volumetric Rendering Methods,
Simultaneous Localization and Mapping, Graph Representations, Image Processing

Experience

Postdoctoral Researcher	Oct 2025–present
- <i>Mitsubishi Electric Research Labs, Cambridge, MA, USA</i>	
Software Developer	Jul 2017–Jul 2018
- <i>AMDOCS Development Center India LLP, Pune, India</i>	

Education

Ph.D.(Engg.) & M.Tech.(Res.) - Electrical Engineering Indian Institute of Science (IISc), Bengaluru, India Prime Ministers Research Fellow (PMRF), IUPRAI Doctoral Dissertation Awardee 2025 Advisor: Prof. Venu Madhav Govindu Thesis: Global Methods for Camera Motion Estimation	2018–2025
B.Tech. - Electronics and Communication Engineering (ECE) National Institute of Technology (NIT), Silchar, India Silver Medalist (ECE Branch Topper)	2013–2017

Projects

- Graph sparsification and disambiguation in structure-from-motion Developing methods to prune input graphs for faster and accurate 3D reconstructions	Aug 2023–present
- Motion averaging in 3D reconstruction problems Developing methods for camera motion estimation in structure-from-motion	Aug 2019–present
- Restoration of images corrupted by various noises using fuzzy based approaches Developed different methods for impulse noise removal from colour images	Jul 2016–May 2017
- K-Map Windows app Developed an application to solve 4 variable K-Map in Sum-of-Product (SOP) form	Mar 2015

Research Papers

Conferences

- **L Manam**, V M Govindu, “[Parallel rigidity matters for bundle adjustment](#),” Conference on Computer Vision and Pattern Recognition (CVPR), 2026 (highlight, 3.6% acceptance rate)
- V Piedade, **L Manam**, M Yamazaki, P Miraldo, “[Revisiting monocular SLAM with spatio-temporal scene modeling](#),” Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- **L Manam**, V M Govindu, “[Leveraging camera triplets for efficient and accurate structure-from-motion](#),” Conference on Computer Vision and Pattern Recognition (CVPR), 2024
- **L Manam**, V M Govindu, “[Fusing directions and displacements in translation averaging](#),” International Conference on 3D Vision (3DV), 2024 (oral presentation, 6% acceptance rate)
- **L Manam**, V M Govindu, “[Sensitivity in translation averaging](#),” Neural Information Processing Systems (NeurIPS), 2023
- C Sidhartha, **L Manam**, V M Govindu, “[Adaptive annealing for robust geometric estimation](#),” Conference on Computer Vision and Pattern Recognition (CVPR), 2023
- **L Manam**, V M Govindu, “[Correspondence reweighted translation averaging](#),” European Conference on Computer Vision (ECCV), 2022
- **L Manam**, A Roy, R H Laskar and F A Talukdar, “[Removal of fixed valued impulse noise using global noise statistics based adaptive histogram fuzzy filter](#),” TENCON 2017 - IEEE Region 10 Conference, 2017

Journals

- **L Manam**, V M Govindu, “[Sensitivity analysis in translation averaging](#),” IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026

- **L Manam**, V M Govindu, “[Unifying viewgraph sparsification and disambiguation of repeated structures in structure-from-motion](#),” International Journal of Computer Vision (IJCV), 2026
- A Roy, **L Manam**, R H Laskar, “[Removal of ‘Salt & Pepper’ noise from color images using adaptive fuzzy technique based on histogram estimation](#),” Multimedia Tools and Applications, 2020
- A Roy, **L Manam** and R H Laskar, “[Region adaptive fuzzy filter: an approach for removal of random valued impulse noise](#),” IEEE Transactions on Industrial Electronics (TIE), 2018
- A Roy, J Singha, **L Manam**, R H Laskar, “[Combination of adaptive vector median filter and weighted mean filter for removal of high density impulse noise from color images](#),” IET Image Processing, 2017

Honours and Awards

- Recipient of [IUPRAI Doctoral Dissertation Award 2025](#) for my Ph.D. thesis (Dec 2025)
- Selected at ICCV 2025 Doctoral Consortium to present my Ph.D. thesis work (Sep 2025)
- Received Kotak-IISc AL-ML Travel Grant to attend CVPR 2024 (Jun 2024)
- Received Pratiksha Trust Travel Fellowship to attend 3DV 2024 (Mar 2024)
- Received Google Travel Grant and NeurIPS 2023 Scholar Award to attend NeurIPS 2023 (Dec 2023)
- Granted Prime Ministers Research Fellowship (from Govt. of India) to support my Ph.D. (Aug 2018)
(among 6.98% of interviewed candidates)
- Received Academic Excellence Award and a Silver Medal at NIT Silchar for scoring the highest GPA (May 2017)
- Awarded Best Volunteer 2015 in Administration at Children of Hope India (Silchar based NGO) (Feb 2016)
- Invited for Dewang Mehta felicitation of engineering students at Kaziranga University Jorhat (Aug 2015)
- Received Academic Excellence Award and a Silver Medal at Delhi Public School Dhanbad (Jul 2013)

Teaching Assistantships

- *Indian Institute of Science, Bengaluru, India*
 - *Course:* Computer Vision (E1-216) Spring 2021-2024
Instructor: Prof. Venu Madhav Govindu
 - *Course:* Stochastic Models and Applications (E1-222) Fall 2020
Instructor: Prof. P S Sastry
- *NPTEL Online Courses*
 - *Course:* Computer Vision and Image Processing (NOC23-EE39) Spring 2023
Instructor: Prof. M K Bhuyan, IIT Guwahati, India
 - *Course:* Computer Vision (NOC22-CS89) Fall 2022
Instructor: Prof. Jayanta Mukhopadhyay, IIT Kharagpur, India

Skills & Technologies

- C, C++, Python
- $\text{\LaTeX}$, MATLAB